

System Acceptance Review
BRACU Mongol-Tori
BRAC University
University Rover Challenge - 2024



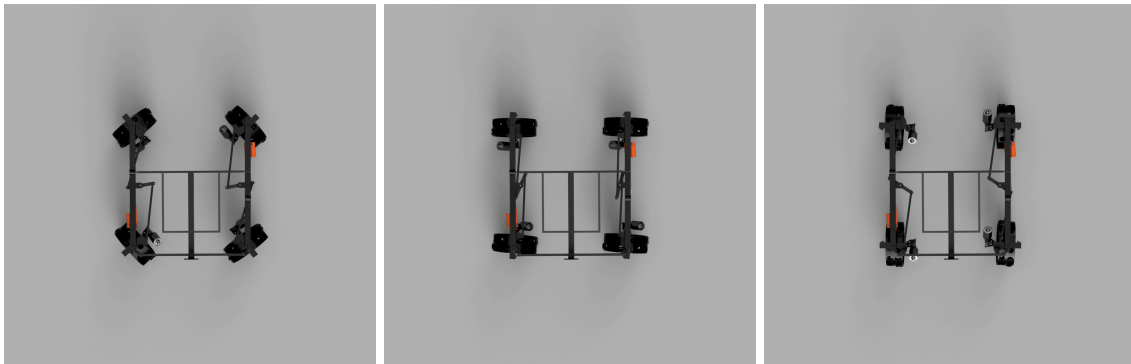
Team Lead: Al Mahir Ahmed
Email: al.mahir.ahmed@g.bracu.ac.bd
Optional Contact:
Name: MD Sanzim Rahman Khan
Email: sanzim.rahman.khan@g.bracu.ac.bd

1. Introduction: For URC 2024, our team proudly presents the redesigned rover, 'Mongol-Tori Phoenix,' paying homage to its namesake with renewed vigor and innovative enhancements. Our team of over 30 passionate undergraduates from multiple disciplines is divided into 5 sub teams: Science, Automation, Arm, Drive and Navigation. This SAR report delves into the details of our rover's core systems, outlining our approach to conquering the Competition Tasks. Additionally, we provide insights into the testing protocols devised to ensure the resilience and reliability of our rover, both in the present and future endeavors. Furthermore, we present our Gantt Chart and budget list that delineates our project timeline and displays the expenses incurred for the formation of our rover and finally, an in-depth exploration of our plans for tackling the Science Task.

2. Core Rover Systems: In this report section, we will look at the various subsystems comprising our rover. Specifically, we will focus on the following key areas: Mechanical, Control, Electronics, Communication, and Science (which will be detailed on page 6).

2.1. Mechanical: After URC 2023, our rover has brought forth significant changes to its mechanical setup. Seeing as the 4 wheeler rocker suspension has worked well, we have kept that while implementing a new arm and omni wheels. We are using aluminum and ebonite to create the new body of the rover, contributing to our weight reduction and rigidity goal.

2.1.1. Omni Wheels: Our rover's mobility stems from its innovative wheel configuration, allowing for omnidirectional movement with adjustable angles at 0° , 45° , and 90° using heavy duty linear actuators. This design enables skid steering motion when the wheels are set at 0° , and facilitates seamless rotation on the spot in both clockwise and anticlockwise directions at 45° . Moreover, at 90° , the rover possesses the capability to execute lateral movement. By positioning the wheels at varying angles (as shown in Figure 1.a.i. below), we have engineered a versatile platform capable of navigating diverse terrains and overcoming challenging obstacles with agility.



Omni Wheels

2.1.2. Modular Systems: Our rover has separate modules dedicated to science, navigation functions, network connectivity, and electronics. This design allows seamless plug-and-play integration of these systems, facilitating customization and swift deployment for diverse mission requirements. By separating functionalities into distinct modules, we mitigate the risk of system-wide cascading failures, ensuring resilience in the face of unforeseen challenges. This approach also minimizes troubleshooting time and enhances overall system stability. Moreover, due to the system's compactness, the rover's body could be sized down for different missions, resulting in a significant weight reduction.

2.1.3. Arm: Learning from the past we have taken a different approach to designing our new robotic manipulator focussing more on optimization. The new arm has 5 degrees of freedom with the 2 degrees involved in positioning the end effector module, 2 degrees for the orientation of the final end effector

and the final degree in charge of the position of the final end effector. The end effector module contains the final end effector, a linear actuator for precise linear positioning of the end effector and a revolute joint for its orientation. The two motors in charge of positioning the end effector module are located near the base of the arm and finally there is one more linear actuator located at the base that orients the end effector module. This design, having most of the arm's weight concentrated at the base, allows for a greater payload to manipulator weight ratio giving a significant payload strength of up to 7 kgs. The 2 finger end effector paired with a laser and solenoid allows for precise manipulation. Moreover, the modular design of the arm helps for seamless removal and replacement for different missions.

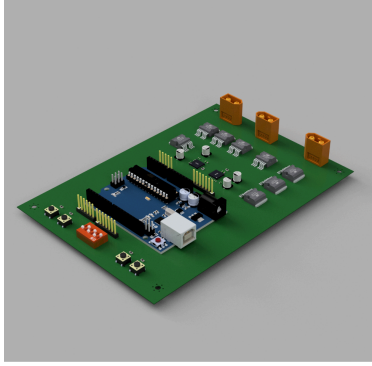
- **Science Arm:** A dedicated arm with two degrees of freedom and an excavator claw has been designed for soil collection during science missions.

2.1.4. Antenna Bending: The omnidirectional antenna mounted on the rover exceeded the allowed height limit. To rectify this, we introduced a new antenna extension mechanism, which enables orientation adjustments directly from the antenna mount's base.

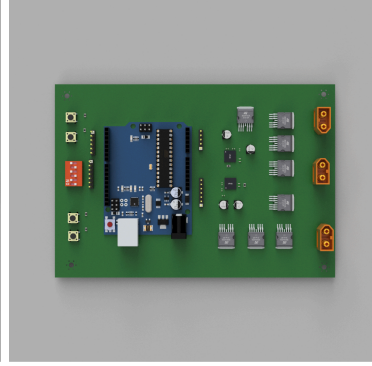
2.2 Control Systems

2.2.1. Software: Our main on board computer features an Intel NUC where we have used ROS Noetic (operating on Ubuntu 20.04) as the backbone of our entire system for streamlined and efficient data transmission between modules. The ROS master node is situated on the rover allowing the rover to make decisions in case communications get lost. Complementing this, our user-friendly web graphical user interface (GUI), intricately linked to ROS, displays rover position and orientation data on an offline map. Along with this a separate science web page displays data curated from different sensors from ROS topics. Battery level data is also displayed on every web page. For modularity different nodes are created to transmit and receive data from different subsystems making sure no system is dependent on another. The use of omni-wheels enables our rovers to execute reliable lateral and translational motions, underscoring our commitment to versatility and maneuverability. Furthermore, our emphasis on intuitive control systems, with the use of gamepads and joysticks, ensures accessibility for users of all skill levels, enhancing usability and reducing operational barriers. Incorporating high-end AI models further enhances our system's capabilities, necessitating a mini PC such as the Intel NUC to support advanced computation. A backup web socket system has been made that works with the web GUI in case of failure from ROS.

2.2.2. Electronics: The electronics subteam prioritized modularity, efficiency, and resilience. Our redesign includes the development of two custom printed circuit boards (PCBs) dedicated to the arm and wheel. Each PCB comprises N channel MOSFET H bridge driven by DRV8702 and relays for motor control both bidirectional and unidirectional. The PCBs have buttons on board for better debugging and manual control. An added feature is the limit switch feedback, used on the arm to turn off the motors and adjust when they reach their limit in its range of motion, and on the omni wheels to make sure the rover base can be automatically set to different modes. Each PCB, housing a microcontroller (Arduino), directly communicates with the central computer (Intel NUC) serially, promoting modular functionality and minimizing cross-system faults. Backup circuits utilizing relays and readily available motor drivers have been integrated to enhance reliability. The rover operates with dual 4s 16V LiPo batteries, each serving distinct functions: one powers the communications and computer, while the other exclusively powers the actuators, sensors and camera modules. The power distribution board (PDB), equipped with step-up and step-down transformers or buck-boost converters tailored to the system's demands, is central to our power management. The PDB oversees power distribution and incorporates safety measures, monitored states, and safety mechanisms. We have integrated a Solid State Relay (SSR) with a killswitch to control power delivery, ensuring safe operation. If communication is interrupted, another SSR serves to reduce locomotive functions.



PCB



PCB (top view)



5 DOF Arm

2.3. Communication: This year we used 3 layers of communication system for gaining uninterrupted control and feed from the rover. The modified and upgraded communication system of this year includes two dedicated point to point network topology that allows the rover to communicate via 2.4GHz and 5.8GHz. The base station integrates two directional Ubiquiti Rocket M2 and Ubiquiti Rocket M5 units, with the WAVLINK router on the rover connecting point-to-point with the Ubiquiti Rocket M2. We used both bands simultaneously as well as independently so that we can maintain high-res video feed from the rover and reliable communication as failsafe. Our communication system itself can detect noise from its surroundings and binds up to a better channel for keeping itself away from any kind of interference that could hinder vision. A rotator paired with an elevation azimuth controller is being used in the base station to maintain line of sight communications with the rover. We used multiple webcams strategically positioned on the rover assigning each of them an IP (thereby converting them into cheap IP cams). This wide vision gave significant improvement in autonomous movement of the rover. Furthermore, if any of these fails to communicate with the base station, we have multiple FPV cams that will still remain alive and can take the control of the rover to complete any expedition. The use of drone assistance to extend communication range is currently being evaluated via a repeater.

3. Approach to Competition Mission:

3.1 Extreme Delivery Mission: Our rover incorporates a robust 4 wheeled rocker bogie suspension system, ensuring reliable mobility across various natural terrains. From soft sands to rugged rocky landscapes, gravel paths, and boulder fields, this suspension system guarantees stability and agility, enabling the rover to overcome obstacles and navigate challenging terrain up to 1 km from the starting point. Moreover, it has passed a 1m drop test, underscoring its durability and resilience in harsh environments. An intuitive web graphical user interface (GUI) designed for effortless control and monitoring of operations. Operators can easily navigate marked paths, conduct visual searches, interpret signs, and navigate to designated pickup/delivery locations using provided GNSS coordinates enabling smooth control. The rover displays its position and orientation data on the offline map, aiding operators in accurately tracking its movements across diverse terrains. This functionality is essential for precisely navigating designated paths and reaching specific coordinates. Equipped with fossil, rock and tool detection models, our rover can visually identify geological features of interest, assisting operators with geological expertise in analyzing ground samples as well as tools hidden from the operators point of view. This capability enables the rover to contribute to scientific exploration by cataloging unique geological specimens using only its onboard camera.

3.2 Equipment Servicing Mission: The omni wheels on our rover enable it to perform holonomic/strafing motions, significantly enhancing maneuverability across intricate terrains and efficient equipment access. This feature ensures reliable movement, crucial for tasks like transporting the cache container to

the lander and maneuvering controls on the lander's panel. The rover's arm boasts a lightweight design with mass concentrated at the base, optimizing the payload-to-arm weight ratio for versatile task handling with minimal energy consumption. Equipped with prismatic joints on the end effector, the arm facilitates precise translational movement along any single arbitrary axis, ideal for tasks such as inserting the cache container into the lander's drawer and manipulating switches or buttons. Leveraging feedback sensors in the rover's arm allows precise control and feedback during task execution. We are employing inverse kinematics to allow the arm to make large movements thereby giving the operator to make more precise and intricate movements later on. This is crucial for tasks requiring manipulation, such as typing commands, directing an antenna with a joystick, and inserting a USB memory stick into the lander's slot.

3.3 Autonomous Navigation Mission: Our rover utilizes GNSS for precise localization and ROS Navigation Stack for autonomous path planning. The system includes a global planner for creating paths between GNSS coordinates and a local planner for short-distance navigation and obstacle avoidance using a camera feed. The SBG Ellipse-D INS and differential GNSS ensure precise positioning with a 1-meter single-point positioning precision and 0.2-degree heading precision. Using OpenCV, our rover detects ArUco tags and enters a holding pattern defined by four GNSS coordinates to locate posts. Utilizing its depth camera, the rover approaches visual markers accurately, stopping at the required distance for completion. Equipped with the YOLO v8 object detection model which is further trained on larger datasets, our rover identifies objects like the rubber mallet and water bottle. A separate ArUco detection model is used to detect markers from long ranges. By leveraging depth perception from its camera, the rover navigates around obstacles to reach object coordinates, halting at the specified distance for task completion. The navigation algorithm incorporates search patterns during visual searches of objects and ArUco markers in case they are not found on first attempts. The web GUI allows operators to place waypoint markers either by clicking on the offline map or manually entering the coordinates onto a search bar. We are currently developing a custom global planner to allow for implementation of our own path planning algorithms.

4. Testing and Operations:

4.1. Sub-system testing: Testing for electronics, omni wheels, manipulator, control, navigation and science began on 1/10/23 and ended on 10/02/24. For the arm, control and navigation, testing was carried out in simulation before testing in the real world. For electronics, veroboard and breadboard circuits were made and simultaneously the PCBs for the respective circuit was being designed iteratively through trial and error of the veroboard and breadboard circuit. The omni wheels were made and tested iteratively through trial and error before arriving at the final version. All versions were tested on rough terrain with rigorous stress testing in terms of weight and payload.

4.2 System testing: Complete bare minimum base systems have been made on the rover and different subsystems have been integrated and tested from 15/01/24 to 28/02/24. These tests were carried out in environments similar to URC. Tests such as drawer opening, turning on switches, long range traversal, point to point navigation and onboard science testing were carried out during this period. These semi mock tests enabled minor developments to be made in terms of quality of life features, especially on the software side.

4.3 Operational testing: Complete operational testing is set to begin on 05/03/24 where we will properly simulate URC conditions with allotted time. In these mock tests we will divide our team into groups of 6 and have each of them complete different missions. The tests will be weekly and regular. We expect to make more developments here in terms of quality of life and may make major developments if need be. This period will continue till 10/05/24 after which we will begin preparations for Utah if we are selected for URC 2024 finals.

5. Science Plan: The modular onboard life detection assay features a custom spectrometer with a UV-Vis laser, spanning wavelengths from 250 nm to 800 nm. This tool enables in-depth analysis of absorption spectra across various biomarkers. With a setup comprising 15 quartz cuvettes, each measuring 12.5 mm x 12.5 mm x 45 mm, ample space is provided for comprehensive sample analysis. A high-quality white screen with low absorption properties ensures precise detection of light diffracted by the cuvettes. A camera then detects the spectra, and Theremino software with our GUI will aid in wavelength peak identification. Beer-Lambert's law then correlates biomarker intensity with concentration, providing a comprehensive understanding of potential life forms.

Among the targeted biomarkers, Flavin Mononucleotide (FMN) holds substantial significance. FMN, playing a pivotal role in metabolic processes, serves as an indicator of potential extant life. Its distinctive absorption peaks at 410 nm, 360 nm, and 450 nm for oxidized and reduced forms enable accurate detection. Additionally, generic extant porphyrin emerges as a versatile biomarker, particularly in aerobic and anaerobic phototrophic organisms. Porphyrins play integral roles within chlorophyll molecules crucial for photosynthesis and are also present in cytochrome structures, facilitating electron transfer in cell membranes. Their preservation in sedimentary rocks underscores their value as indicators of extinct life forms.

Fe-S clusters represent another vital biomarker, suggesting potential alternative metabolic pathways in environments abundant in Ferric oxides reminiscent of the Martian surface. These clusters are essential for autotrophic ATP synthesis and are closely associated with life evolution in challenging environments. Their presence typically implies the existence of proteins, as Fe-S clusters are commonly bound within protein structures. Detecting ammonia, primary or secondary amines, and amino acids provides valuable insights into cellular metabolism. This detection relies on the Ninhydrin test, where the formation of the diketohydrin complex induces a distinct purple colouration upon completion of the reaction with amino acids.

The rover employs advanced sensors to conduct comprehensive assessments for soil analysis and collection. Data transmission, analysis, and display are facilitated through a graphical user interface (GUI), with continuous storage in CSV format for graphical representation. Conductivity and salinity tests are conducted using an EC sensor (Electrical Conductivity meter), enabling direct evaluation of soil composition. Furthermore, an NPK sensor monitors nitrogen (N), phosphorus (P), and potassium (K) levels, which are crucial for understanding soil fertility. The science arm is calibrated with thermal and moisture sensors to provide accurate readings during soil collection and analysis.

The retractable science arm, featuring a downward-oriented claw, extends and retracts using an actuator mechanism. It is adept at efficiently collecting soil from the top 10 cm layer, with further extension enabling retrieval of soil from deeper layers. An excavator claw used in our previous year will be used to extract the sample. Following soil collection, the arm tilts backwards, directing the soil to a roller for crushing, quantifying the amount crushed through rotational speed. This versatile mechanism accommodates various soil types and can even crush small rocks. Collected soil is then directed toward designated cuvettes for analysis.

Moreover, the soil dispenser incorporates a sealed cache collection system to prevent spills and minimize contamination and radiation risks. This system includes a roller with dual rotation, ensuring downward soil disposal without spillage. A tilted blade below the roller guides soil towards the cuvettes, with a tubular hole accommodating a sealed cache box calibrated to hold at least 5 grams of soil sample. The lid securely locks, sealing the tube pathway as it fills, directing excess soil toward the cuvettes for further analysis. This process ensures sample integrity while facilitating the aforementioned spectroscopic analysis of the prepared solution.